

Digital Logic Design Lab

هذا العمل صدقة جارية عن أرواح موتى وشهداء المسلمين عامة وشهداء فلسطين خاصة
اللهم إنهم في ذمتك وحبل جوارك، فقههم من فتنة القبر وعذاب النار، وأنت أهل الوفاء
والحق، اللهم اغفر لهم وارحمهم، فإنك أنت الغفور الرحيم
اللهم اغفر لهم وارحمهم واعف عنهم وعافهم، وأكرم نزلهم ووسع مدخلهم، واغسلهم
بالماء والثلج والبرد، ونقهم من الخطايا كما ينقى الثوب الأبيض من الدنس، وأبدلهم داراً
خيراً من دارهم وأهلاً خيراً من أهلهم وأزواجاً خيراً من أزواجهم، وقهم فتنة القبر وعذاب
النار

اللهم إنهم عبيدك وبناء عبيدك وبناء إماءك كانوا يشهدون أن لا إله إلا أنت، وأن محمداً
عبدك ورسولك، وأنت أعلم بهم، اللهم إن كانوا محسنين فزدهم في إحسانهم، وإن كانوا
مسيئين فتجاوز عن سيئاتهم، اللهم لا تحرمنا أجرهم ولا تفتنا بعدهم

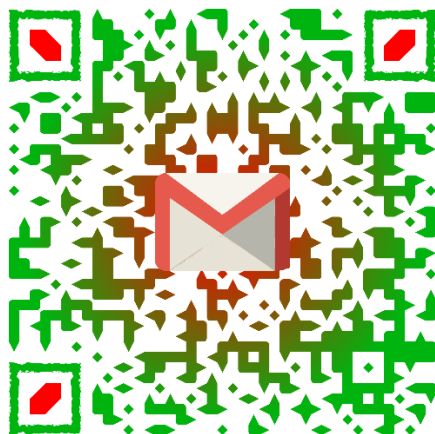
اللهم سد رمي المقاومة
اللهم لا ترفع لليهود راية ولا تحقق لهم غاية واجعلهم لمن خلفهم عبرة وآية

قال تعالى: ﴿سُبْحَانَ الَّذِي أَسْرَى بِعَبْدِهِ لَيْلًا مِنَ الْمَسْجِدِ الْحَرَامِ إِلَى الْمَسْجِدِ الْأَقْصَا الَّذِي
بَارَكْنَا حَوْلَهُ لِنُرِيَهُ مِنْ آيَاتِنَا إِنَّهُ هُوَ السَّمِيعُ الْبَصِيرُ * وَآتَيْنَا مُوسَى الْكِتَابَ وَجَعَلْنَاهُ هُدًى
لِبَنِي إِسْرَائِيلَ أَلَّا تَتَّخِذُوا مِنْ دُونِي وَكِيلًا * ذُرِّيَّةً مِنْ حَمَلْنَا مَعَ نُوحٍ إِنَّهُ كَانَ عَبْدًا شَكُورًا *
وَقَضَيْنَا إِلَى بَنِي إِسْرَائِيلَ فِي الْكِتَابِ لَتُفْسِدُنَّ فِي الْأَرْضِ مَرَّتَيْنِ وَلَتَعْلُنَّ عُلُوقًا كَبِيرًا * فَإِذَا
جَاءَ وَعْدُ أُولَاهُمَا بَعَثْنَا عَلَيْكُمْ عِبَادًا لَنَا أُولِي بَأْسٍ شَدِيدٍ فَجَاسُوا خِلَالَ الدِّيَارِ وَكَانَ وَعْدًا
مَفْعُولًا * ثُمَّ رَدَدْنَا لَكُمُ الْكَرَّةَ عَلَيْهِمْ وَأَمْدَدْنَاكُمْ بِأَمْوَالٍ وَبَنِينَ وَجَعَلْنَاكُمْ أَكْثَرَ نَفِيرًا * إِنْ
أَحْسَنْتُمْ أَحْسَنْتُمْ لِأَنْفُسِكُمْ وَإِنْ أَسَأْتُمْ فَلَهَا فَإِذَا جَاءَ وَعْدُ الْآخِرَةِ لِيَسُوءُوا وُجُوهَكُمْ وَلِيَدْخُلُوا
الْمَسْجِدَ كَمَا دَخَلُوهُ أَوَّلَ مَرَّةٍ وَلِيُتَبِّرُوا مَا عَلَوْا تَتْبِيرًا * عَسَى رَبُّكُمْ أَنْ يَرْحَمَكُمْ وَإِنْ عُذْتُمْ
عُدْنَا وَجَعَلْنَا جَهَنَّمَ لِلْكَافِرِينَ حَصِيرًا﴾ {سورة الإسراء}

ملاحظة مهمة جداً: هذا العمل منسوب لفريق بسواعدنا نبنيها - سبارك ويمنع منعاً باتاً التعديل على الملف بأي شكل من الأشكال ونشره من أي شخص أو جهة للطلاب إلا بصيغته الأصلية هذه والمنشورة من قبل منشأ هذا العمل

والكمال لله وحده لهذا من الممكن أن تتواجد بعض الأخطاء في الحلول لذلك نرجو منكم الاستفسار عن الأسئلة التي تريدون السؤال عنها بالتواصل مع منشأ هذا العمل

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Question 1

Correct

Mark 1.00 out of 1.00

Flag question

What additional feature have a sequential circuit, compared to a combinational circuit?

Select one:

- ☐ a. AND or OR gates
- ☐ b. a power transistor
- ☒ c. a memory state ✓
- ☐ d. clock pulse generators

The correct answer is: a memory state

Question 2

Incorrect

Mark 0.00 out of 2.00

Flag question

How many represents a number of output lines for a decoder with 4 input lines

Answer: 2



The correct answer is: 16

Question 3

Correct

Mark 1.00 out of 1.00

Flag question

When happens a state chngement in a clocked sequential circuit?

Select one:

- ☐ a. when the clock signal is high
- ☒ b. on the rising or falling edges of the clock ✓
- ☐ c. at any moment: it depends on the inputs
- ☐ d. during the period "T" of the clock

The correct answer is: on the rising or falling edges of the clock

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

Which logic gates can be used to make a flip-flop?

Select one:

- ☐ a. OR
- ☐ b. EXCLUSIVE OR
- ☒ c. 1 NAND ✓
- ☐ d. AND

The correct answer is: 1 NAND

Question 5

Incorrect

Mark 0.00 out of 1.00

Flag question

In a coder with priority (Priority Encoder):

Select one:

- ☐ a. only the input activated first is considered active
- ☐ b. when the inputs are active, the outputs are not
- ☒ c. the output is correct only if the inputs are active one at a time ✖
- ☐ d. the input with heavier (or lighter) weight is considered active

The correct answer is: the input with heavier (or lighter) weight is considered active

Question 6

Incorrect

Mark 0.00 out of 1.00

Flag question

All numbers from 0 to 15 are represented in a BCD-7-segment decoder driver.

Select one:

- ☒ a. True ✖
- ☐ b. the question wrong
- ☐ c. no answer
- ☐ d. False

The correct answer is: False

Question 7

Incorrect

Mark 0.00 out of 1.00

Flag question

What is a sequential circuit?

Select one:

- ☐ a. a circuit composed of a sequence of elementary logic gates, e.g. AND gates
- ☒ b. a circuit in which the outputs depend only on the inputs at any moment **✗**
- ☐ c. a loop circuit, always repeating the same cycle
- ☐ d. a circuit containing memory elements, in which the outputs not only depend on the input but also on the state of the memory element.

The correct answer is: a circuit containing memory elements, in which the outputs not only depend on the input but also on the state of the memory element.

Question 8

Incorrect

Mark 0.00 out of 1.00

Flag question

Which kind of display is it better to use for the display of the alphanumeric characters?

Select one:

- ☐ a. dot-matrix display (e.g. 6 x 8)
- ☐ b. 7-segment display
- ☒ c. nixie display **✗**
- ☐ d. 14-16-18-segment display

The correct answer is: 14-16-18-segment display

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Question 9

Incorrect

Mark 0.00 out of
1.00

Flag question

AND,OR, NAND and NOR considered as sequential circuit ✖

Your answer is incorrect.

The correct answer is:

AND,OR, NAND and NOR considered as[combinational circuits]

Question 10

Incorrect

Mark 0.00 out of
1.00

Flag question

Which kind of code must be supplied to the inputs of a BCD-to-decimal coder?

Select one:

- ☐ a. Gray
- ☐ b. pure binary
- ☒ c. excess 3 ✖
- ☐ d. BCD

The correct answer is: BCD

Question 11

Incorrect

Mark 0.00 out of
1.00

Flag question

How is numerical data displayed normally, in digital systems?

Select one:

- ☐ a. with a 7-segment display
- ☒ b. with a speaker ❌
- ☐ c. with a multimeter
- ☐ d. with a counter

The correct answer is: with a 7-segment display

Question 12

Incorrect

Mark 0.00 out of
1.00

Flag question

How many inputs and outputs has a decimal to BCD encoder without priority?

Select one:

- ☐ a. 9 inputs and 3 outputs
- ☐ b. 9 or 10 inputs and 4 outputs
- ☒ c. 8 inputs and 4 outputs ❌
- ☐ d. 10 inputs and 3 outputs

The correct answer is: 9 or 10 inputs and 4 outputs

Question 13

Correct

Mark 1.00 out of 1.00

Flag question

An RS flip-flop is the basic type of sequential logic circuits. ✓

Your answer is correct.

The correct answer is:

An RS flip-flop is the basic type of [sequential logic circuits.]

Question 14

Correct

Mark 1.00 out of 1.00

Flag question

In a 7-segment display decoder driver using negative logic (common anode display), a low output level represents a segment in the ON state.

Select one:

- ☐ a. No answer
- ☒ b. True ✓
- ☐ c. False
- ☐ d. the question wrong

The correct answer is: True

Question 15

Incorrect

Mark 0.00 out of
1.00

Flag question

What is the function of a decimal-binary encoder?

Select one:

- ☐ a. to convert from decimal to binary code
- ☐ b. to convert from binary to decimal code
- ☒ c. to convert an analog signal into a digital one ✖
- ☐ d. to convert a BCD number into an hexadecimal one

The correct answer is: to convert from decimal to binary code

Question 16

Correct

Mark 1.00 out of
1.00

Flag question

What do you mean, when you say that a flip-flop is an indeterminate state?

Select one:

- ☐ a. that the flip-flop tends to oscillate indefinitely
- ☐ b. that the outputs are not fired
- ☒ c. that after switching flip-flop, the memory state is undefined, and can be $Q=0$ and $???=1$ or $Q=1$ and $???=0$ ✔
- ☐ d. that the outputs are not at level 0, nor 1

The correct answer is: that after switching flip-flop, the memory state is undefined, and can be $Q=0$ and $???=1$ or $Q=1$ and $???=0$

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Question 17

Incorrect

Mark 0.00 out of 2.00

Flag question

How many flip flop we have?

Answer: 3



The correct answer is: 4

Question 18

Incorrect

Mark 0.00 out of 1.00

Flag question

a 7-segment display shows:

Select one:

- ☐ a. all the letters of the alphabet
- ☒ b. all alphanumeric characters
- ☐ c. all numbers from 0 to 9, the letters A, b, c, d, E, F and some other symbols
- ☐ d. only the numbers from 0 to 9

The correct answer is: all numbers from 0 to 9, the letters A, b, c, d, E, F and some other symbols

Question 19

Not answered

Marked out of 2.00

Flag question

With common cathode, driven with

Answer:



The correct answer is: positive logic

Question 20

Incorrect

Mark 0.00 out of
1.00

Flag question

Which is the better type of drive for an led display?

Select one:

- ☐ a. Current
- ☒ b. R ✖
- ☐ c. Voltage
- ☐ d. Micro

The correct answer is: Current

Question 21

Incorrect

Mark 0.00 out of
1.00

Flag question

A binary-decimal coder is composed of:

Select one:

- ☐ a. logic gates only
- ☐ b. logic gates and flip-flops
- ☒ c. flip-flops only ✖
- ☐ d. none of them

The correct answer is: logic gates only

Question 22

Incorrect

Mark 0.00 out of
1.00

Flag question

What is the function of a binary-decimal coder?

Select one:

- ☐ a. to input data in binary way into a digital system
- ☒ b. to input data in binary way into a Analog system ❌
- ☐ c. to convert a binary number into decimal one
- ☐ d. to count

The correct answer is: to convert a binary number into decimal one

Question 23

Incorrect

Mark 0.00 out of
1.00

Flag question

How many inputs and outputs has a BCD-decimal coder SN74LS42 got?

Select one:

- ☐ a. 4 inputs and 10 outputs
- ☐ b. 10 inputs and 4 outputs
- ☒ c. 4 inputs and 9 outputs ❌
- ☐ d. 9 inputs and 4 outputs

The correct answer is: 4 inputs and 10 outputs

Question 1

Incorrect

Mark 0.00 out of
1.50

Flag question

If the input J is connected through K input of J-K, then flip-flop will behave as a

Select one:

- ☒ a. Master slave JK flip-flop ❌
- ☐ b. T type flip-flop
- ☐ c. D type flip-flop
- ☐ d. S-R flip-flop

The correct answer is: T type flip-flop

Question 2

Correct

Mark 1.50 out of
1.50

Flag question

The digital multiplexer is basically a combination logic circuit to perform the operation

Select one:

- ☐ a. OR-AND
- ☒ b. AND-OR ✔️
- ☐ c. AND-AND
- ☐ d. OR-OR

The correct answer is: AND-OR

Question 3

Incorrect

Mark 0.00 out of
1.50

Flag question

Given $F1(A,B,C,D) = \text{SOP}(0, 2, 4, 8)$, $F2(A,B,C,D) = \text{SOP}(1, 2, 3, 4, 15)$ and $F3 = F1 + F2$ then:

Select one:

- ☐ a. $F3(A,B,C,D) = \text{SOP}(0, 1, 2, 3, 4, 8, 15)$
- ☐ b. $F3(A,B,C,D) = \text{SOP}(0, 1, 2, 3, 4, 8)$
- ☒ c. $F3(A,B,C,D) = \text{SOP}(2, 4)$ ✖
- ☐ d. $F3(A,B,C,D) = \text{SOP}(0, 1, 2, 3, 4)$

The correct answer is: $F3(A,B,C,D) = \text{SOP}(0, 1, 2, 3, 4, 8, 15)$

Question 4

Correct

Mark 1.50 out of
1.50

Flag question

An OR gate can be imagined as

Select one:

- ☒ a. Switches connected in parallel ✔
- ☐ b. None of these
- ☐ c. MOS transistors connected in series
- ☐ d. Switches connected in series

The correct answer is: Switches connected in parallel

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Question 5

Correct

Mark 1.50 out of
1.50

Flag question

The EXCLUSIVE NOR gate is equivalent to which gate followed by an inverter ?

Select one:

- ☐ a. OR
- ☐ b. AND
- ☐ c. NAND
- ☒ d. XOR ✓

The correct answer is: XOR

Question 6

Incorrect

Mark 0.00 out of
1.50

Flag question

A combinational logic circuit which generates a particular binary word or number is

Select one:

- ☐ a. Decoder
- ☐ b. Encoder
- ☐ c. Demultiplexer
- ☒ d. Multiplexer ✗

The correct answer is: Decoder

Question 7

Correct

Mark 2.50 out of 2.50

Flag question

Which of the following gates are added to the inputs of the OR gate to convert it to the NAND gate ?

Select one:

- ☐ a. OR
- ☐ b. XOR
- ☒ c. NOT ✓
- ☐ d. AND

The correct answer is: NOT

Question 8

Incorrect

Mark 0.00 out of 1.50

Flag question

For which of the following flip-flops, the output is clearly defined for all combinations of two inputs ?

Select one:

- ☐ a. D flip-flop
- ☐ b. J-K flip-flop
- ☐ c. R-S flip-flop
- ☒ d. Q type flip-flop ✗

The correct answer is: J-K flip-flop

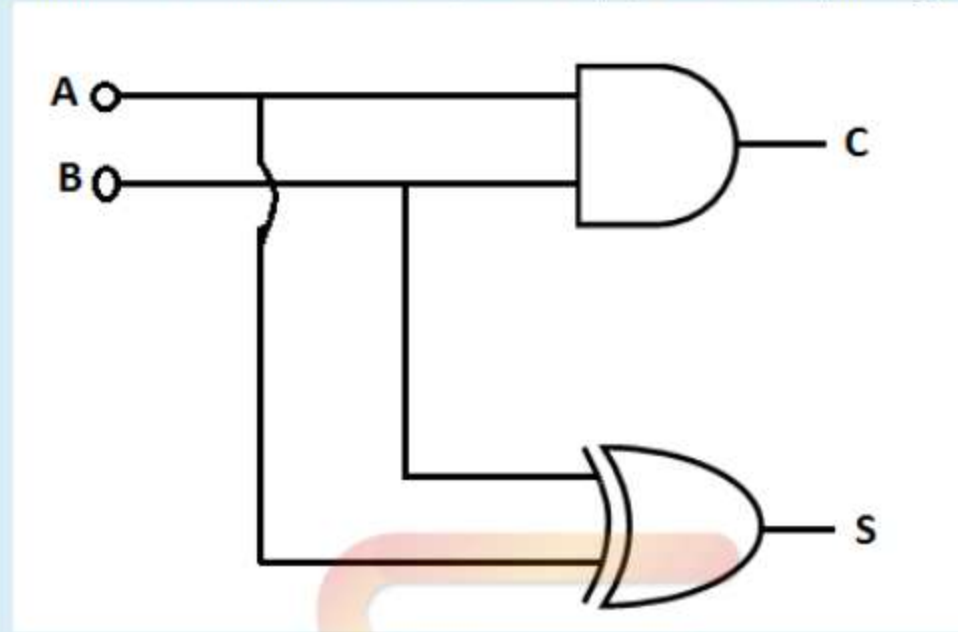
Question 9

Incorrect

Mark 0.00 out of
2.50

Flag question

For the circuit shown for $AB = 00$, $AB = 01$, C, S values respectively are



Select one:

- ☐ a. 0, 0 and 0, 1
- ☐ b. 0, 0 and 1, 0
- ☒ c. 0, 1 and 0, 0 ✖
- ☐ d. 1, 0 and 0, 0

Your answer is incorrect.

The correct answer is: 0, 0 and 0, 1

Question 10

Incorrect

Mark 0.00 out of
1.50

Flag question

A demultiplexer is used to

Select one:

- ☐ a. Both route the data and perform serial to parallel
- ☐ b. Perform serial to parallel conversion
- ☒ c. Route the data from single input to one of many outputs ✖
- ☐ d. Select data from several inputs and route it to single output

The correct answer is: Both route the data and perform serial to parallel

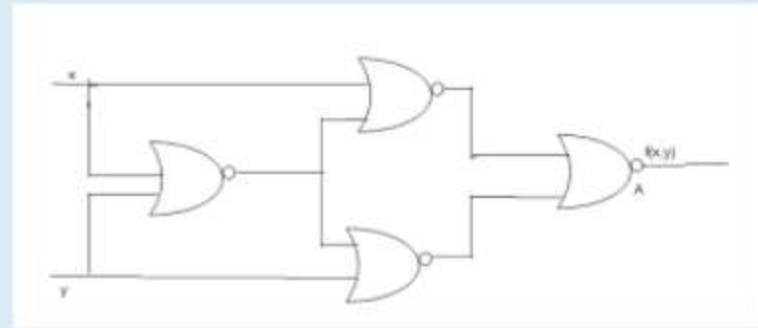
Question 11

Correct

Mark 2.50 out of 2.50

Flag question

Identify the logic function performed by the circuit shown in the given figure



Select one:

- ☒ a. Exclusive NOR ✓
- ☐ b. NOR
- ☐ c. NAND
- ☐ d. Exclusive OR

Your answer is correct.

The correct answer is: Exclusive NOR

Question 12

Incorrect

Mark 0.00 out of 1.50

Flag question

To implement the Boolean function $F = A \oplus B \oplus C$ we can use a 3-to-8 decoder and connect:

Select one:

- ☐ a. 4 different lines from its output to an OR gate
- ☐ b. 5 different lines from its output to an OR gate
- ☒ c. 3 different lines from its output to an OR gate ✗
- ☐ d. 6 different lines from its output to an OR gate

The correct answer is: 4 different lines from its output to an OR gate

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Question 13

Correct

Mark 1.50 out of 1.50

Flag question

The function of a multiplexer is

Select one:

- ☐ a. to decode information
- ☐ b. to transit data on N lines
- ☒ c. to select 1 out of N input data sources and to transmit it to single channel ✓
- ☐ d. to perform serial to parallel conversion

The correct answer is: to select 1 out of N input data sources and to transmit it to single channel

Question 14

Correct

Mark 1.50 out of 1.50

Flag question

How many flip-flop are needed to divide the input frequency by 64 ?

Select one:

- ☐ a. 8
- ☒ b. 6 ✓
- ☐ c. 2
- ☐ d. 5

The correct answer is: 6

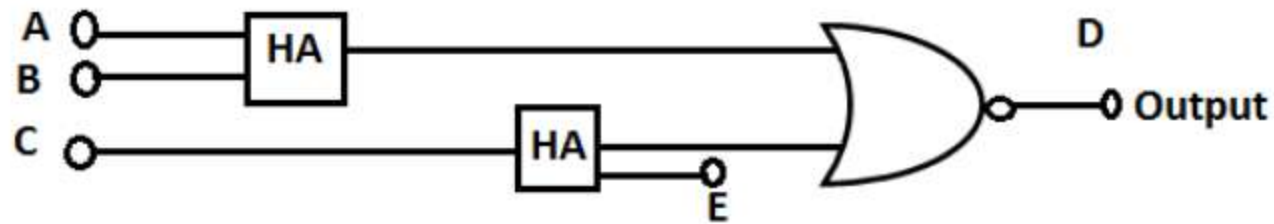
Question 15

Correct

Mark 2.50 out of 2.50

Flag question

The circuit shown below is the



Select one:

- ☒ a. Full adder ✓
- ☐ b. None of these
- ☐ c. Parity checker
- ☐ d. Full subtractor

Your answer is correct.

The correct answer is: Full adder

Question 16

Incorrect

Mark 0.00 out of 1.50

Flag question

Which of the following adders can add two binary numbers at a time ?

Select one:

- ☐ a. Carry-save-adder
- ☐ b. Carry-look-ahead adder
- ☐ c. Full adder
- ☒ d. Parallel adder ✗

The correct answer is: Full adder

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Question 17

Correct

Mark 1.50 out of 1.50

Flag question

The output of NOR gate is

Select one:

- ☐ a. High if all of its inputs are high
- ☐ b. Low if all of its inputs are low
- ☒ c. High if all of its inputs are low ✓
- ☐ d. High if only of its inputs is low

The correct answer is: High if all of its inputs are low

Question 18

Correct

Mark 1.50 out of 1.50

Flag question

How many lines the truth table for a four-input NOR gate would contain to cover all possible input combinations ?

Select one:

- ☐ a. 12
- ☒ b. 16 ✓
- ☐ c. 8
- ☐ d. 4

The correct answer is: 16

Question 19

Incorrect

Mark 0.00 out of
1.50

Flag question

A stable multivibrator is used as

Select one:

- ☐ a. Voltage to frequency converter
- ☐ b. Demultiplexer
- ☒ c. Frequency to voltage converter ❌
- ☐ d. Comparator circuit

The correct answer is: Comparator circuit

Question 20

Correct

Mark 1.50 out of
1.50

Flag question

The master slave JK flip-flop is effectively a combination of

Select one:

- ☒ a. A SR flip-flop and a T flip-flop ✔️
- ☐ b. A T flip-flop and a D flip-flop
- ☐ c. An SR flip-flop and a D flip-flop
- ☐ d. Two D flip-flops

The correct answer is: A SR flip-flop and a T flip-flop

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Question 21

Correct

Mark 1.50 out of 1.50

Flag question

The flip-flops which operate in synchronism with external clock pulses are known as

Select one:

- ☐ a. Asynchronous flip-flop
- ☐ b. Either of the above
- ☒ c. Synchronous flip-flop ✓
- ☐ d. None of these

The correct answer is: Synchronous flip-flop

Question 22

Incorrect

Mark 0.00 out of 1.50

Flag question

When an inverter is placed between both inputs of an SR flip-flop, then resulting flip-flop is

Select one:

- ☐ a. JK flip-flop
- ☒ b. Master slave JK flip-flop ✗
- ☐ c. SR flip-flop
- ☐ d. D flip-flop

The correct answer is: D flip-flop

Question 23

Correct

Mark 1.50 out of 1.50

Flag question

The number of full and half-adders required to add 16-bit numbers is

Select one:

- ☐ a. 4 half-adders, 12 full-adders
- ☐ b. 16 half-adders, 0 full-adders
- ☒ c. 1 half-adder, 15 full-adders ✓
- ☐ d. half-adders, 8 full-adders

The correct answer is: 1 half-adder, 15 full-adders

Question 24

Correct

Mark 1.50 out of 1.50

Flag question

Adders:

Select one:

- ☐ a. is called so because a full adder involves two half-adders
- ☒ b. All of these ✓
- ☐ c. needs two input and generates two output
- ☐ d. adds 2 bits

The correct answer is: All of these

Question 25

Correct

Mark 1.50 out of 1.50

Flag question

What logic function is produced by adding an inverter to the output of an AND gate ?

Select one:

- ☐ a. OR
- ☒ b. NAND ✓
- ☐ c. NOR
- ☐ d. XOR

The correct answer is: NAND

Question 26

Incorrect

Mark 0.00 out of 1.50

Flag question

How many minimal numbers of basic logic gates (NOT, AND, OR) need to represent the Boolean function ?

Select one:

- ☐ a. Two NOT, one OR.
- ☐ b. One NOT, one AND.
- ☒ c. Two NOTs, one OR, one AND. ✗
- ☐ d. One NOT, one OR.

The correct answer is: One NOT, one OR.

Question 27

Correct

Mark 1.50 out of 1.50

Flag question

What is the largest number of data inputs which a data selector with two control inputs can have ?

Select one:

- ☒ a. 4 ✓
- ☐ b. 16
- ☐ c. 8
- ☐ d. 2

The correct answer is: 4

Question 28

Correct

Mark 2.50 out of 2.50

Flag question

The combinational circuit given below is implemented with two NAND gates. To which of the following individual gates is its equivalent?



Select one:

- ☐ a. XOR
- ☐ b. OR
- ☒ c. AND ✓
- ☐ d. NOT

Your answer is correct.

The correct answer is: AND

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Question 29

Correct

Mark 1.50 out of 1.50

Flag question

Which of the following expressions is not equivalent to X' ?

Select one:

- ☒ a. $X \text{ NOR } 1$ ✓
- ☐ b. $X \text{ NAND } 1$
- ☐ c. $X \text{ NAND } X$
- ☐ d. $X \text{ NOR } X$

The correct answer is: $X \text{ NOR } 1$

Question 30

Correct

Mark 1.50 out of 1.50

Flag question

In a positive edge triggered JK flip-flop, a low J and low K produces

Select one:

- ☐ a. None of these
- ☐ b. Low state
- ☒ c. No change ✓
- ☐ d. High state

The correct answer is: No change

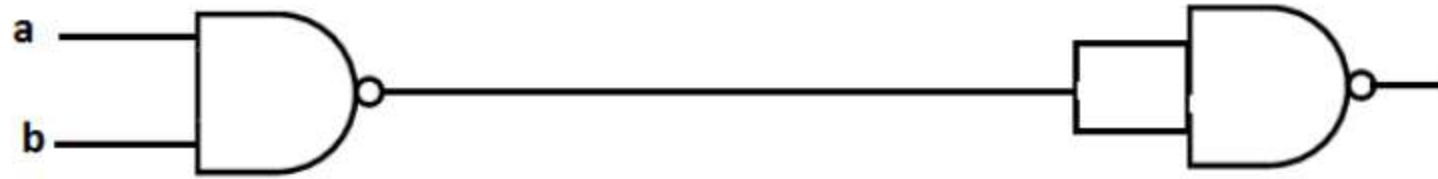
Question 1

Correct

Mark 2.50 out of 2.50

Flag question

The combinational circuit given below is implemented with two NAND gates. To which of the following individual gates is its equivalent?



Select one:

- ☐ a. XOR
- ☐ b. NOT
- ☒ c. AND ✓
- ☐ d. OR

Your answer is correct.

The correct answer is: AND

Question 2

Correct

Mark 1.50 out of 1.50

Flag question

An OR gate can be imagined as

Select one:

- ☒ a. Switches connected in parallel ✓
- ☐ b. None of these
- ☐ c. Switches connected in series
- ☐ d. MOS transistors connected in series

The correct answer is: Switches connected in parallel

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Question 3

Incorrect

Mark 0.00 out of
1.50

Flag question

For which of the following flip-flops, the output is clearly defined for all combinations of two inputs ?

Select one:

- ☐ a. R-S flip-flop
- ☐ b. J-K flip-flop
- ☐ c. Q type flip-flop
- ☒ d. D flip-flop ✖

The correct answer is: J-K flip-flop

Question 4

Correct

Mark 1.50 out of
1.50

Flag question

What is the largest number of data inputs which a data selector with two control inputs can have ?

Select one:

- ☐ a. 2
- ☐ b. 16
- ☒ c. 4 ✔
- ☐ d. 8

The correct answer is: 4

Question 5

Correct

Mark 1.50 out of 1.50

Flag question

The EXCLUSIVE NOR gate is equivalent to which gate followed by an inverter ?

Select one:

- ☐ a. AND
- ☒ b. XOR ✓
- ☐ c. NAND
- ☐ d. OR

The correct answer is: XOR

Question 6

Correct

Mark 2.50 out of 2.50

Flag question

Which of the following gates are added to the inputs of the OR gate to convert it to the NAND gate ?

Select one:

- ☐ a. AND
- ☐ b. OR
- ☒ c. NOT ✓
- ☐ d. XOR

The correct answer is: NOT

Question 7

Correct

Mark 1.50 out of 1.50

Flag question

To implement the Boolean function $F = A \oplus B \oplus C$ we can use a 3-to-8 decoder and connect:

Select one:

- ☐ a. 6 different lines from its output to an OR gate
- ☒ b. 4 different lines from its output to an OR gate ✓
- ☐ c. 5 different lines from its output to an OR gate
- ☐ d. 3 different lines from its output to an OR gate

The correct answer is: 4 different lines from its output to an OR gate

Question 8

Incorrect

Mark 0.00 out of 1.50

Flag question

A demultiplexer is used to

Select one:

- ☒ a. Perform serial to parallel conversion ✗
- ☐ b. Route the data from single input to one of many outputs
- ☐ c. Both route the data and perform serial to parallel
- ☐ d. Select data from several inputs and route it to single output

The correct answer is: Both route the data and perform serial to parallel

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Question 9

Correct

Mark 1.50 out of 1.50

Flag question

The function of a multiplexer is

Select one:

- ☐ a. to perform serial to parallel conversion
- ☐ b. to transit data on N lines
- ☐ c. to decode information
- ☒ d. to select 1 out of N input data sources and to transmit it to single channel ✓

The correct answer is: to select 1 out of N input data sources and to transmit it to single channel

Question 10

Incorrect

Mark 0.00 out of 1.50

Flag question

The digital multiplexer is basically a combination logic circuit to perform the operation

Select one:

- ☐ a. AND-OR
- ☐ b. AND-AND
- ☒ c. OR-OR ✗
- ☐ d. OR-AND

The correct answer is: AND-OR

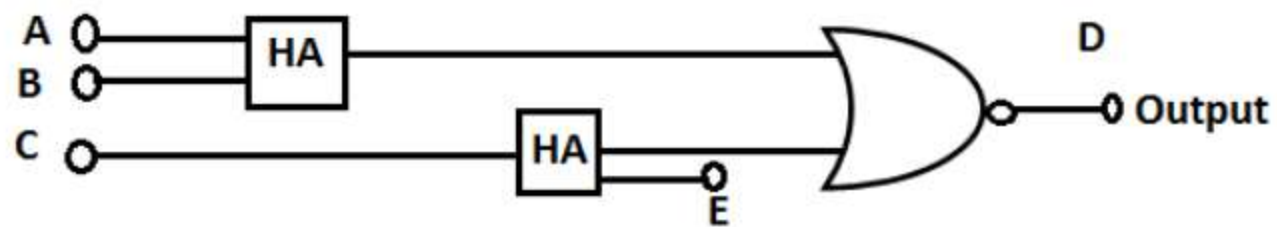
Question 11

Correct

Mark 2.50 out of 2.50

Flag question

The circuit shown below is the



Select one:

- ☐ a. Full subtractor
- ☐ b. None of these
- ☐ c. Parity checker
- ☒ d. Full adder ✓

Your answer is correct.

The correct answer is: Full adder

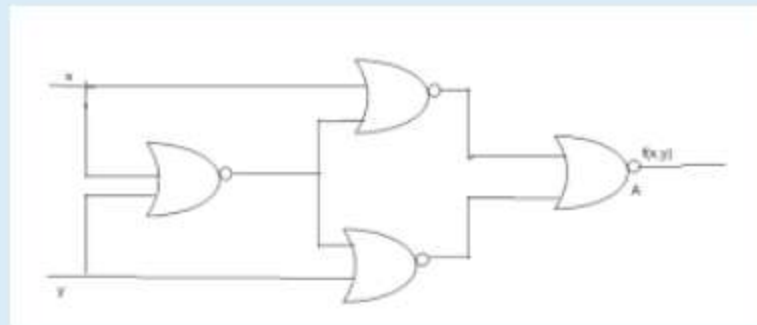
Question 12

Correct

Mark 2.50 out of 2.50

Flag question

Identify the logic function performed by the circuit shown in the given figure



Select one:

- ☐ a. NAND
- ☐ b. NOR
- ☒ c. Exclusive NOR ✓
- ☐ d. Exclusive OR

Your answer is correct.

The correct answer is: Exclusive NOR

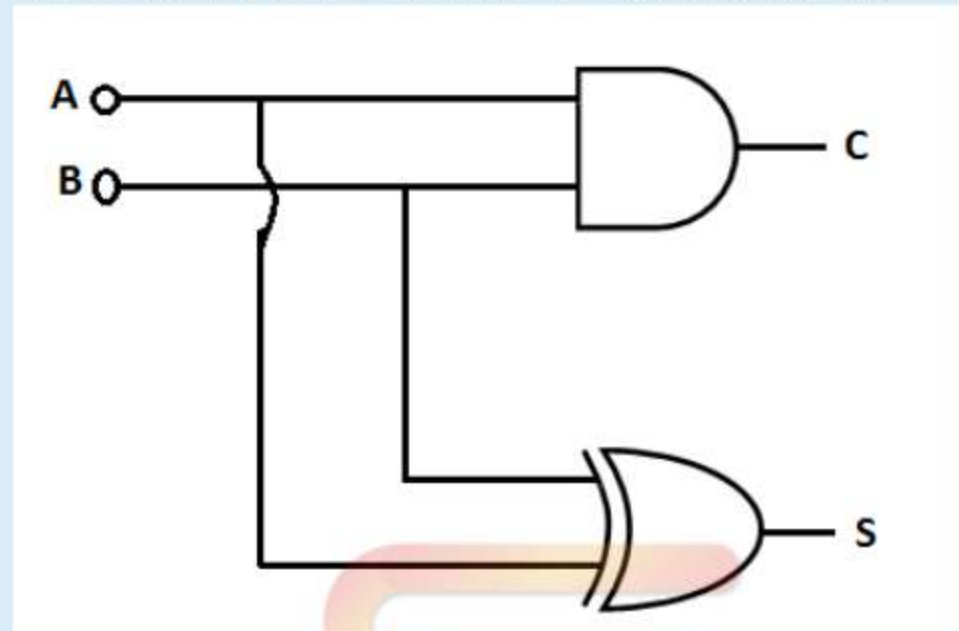
Question 13

Correct

Mark 2.50 out of 2.50

Flag question

For the circuit shown for $AB = 00$, $AB = 01$, C, S values respectively are



Select one:

- ☐ a. 0, 1 and 0, 0
- ☐ b. 1, 0 and 0, 0
- ☒ c. 0, 0 and 0, 1 ✓
- ☐ d. 0, 0 and 1, 0

Your answer is correct.

The correct answer is: 0, 0 and 0, 1

Question 14

Incorrect

Mark 0.00 out of
1.50

Flag question

If the input J is connected through K input of J-K, then flip-flop will behave as a

Select one:

- ☐ a. T type flip-flop
- ☐ b. Master slave JK flip-flop
- ☒ c. D type flip-flop ✖
- ☐ d. S-R flip-flop

The correct answer is: T type flip-flop

Question 15

Correct

Mark 1.50 out of
1.50

Flag question

Which of the following expressions is not equivalent to X' ?

Select one:

- ☐ a. $X \text{ NOR } X$
- ☐ b. $X \text{ NAND } X$
- ☒ c. $X \text{ NOR } 1$ ✔
- ☐ d. $X \text{ NAND } 1$

The correct answer is: $X \text{ NOR } 1$

Question 16

Correct

Mark 1.50 out of 1.50

Flag question

Which of the following adders can add two binary numbers at a time ?

Select one:

- ☐ a. Carry-look-ahead adder
- ☐ b. Parallel adder
- ☒ c. Full adder ✓
- ☐ d. Carry-save-adder

The correct answer is: Full adder

Question 17

Incorrect

Mark 0.00 out of 1.50

Flag question

A combinational logic circuit which generates a particular binary word or number is

Select one:

- ☒ a. Encoder ✗
- ☐ b. Multiplexer
- ☐ c. Decoder
- ☐ d. Demultiplexer

The correct answer is: Decoder

Question 18

Correct

Mark 1.50 out of 1.50

Flag question

The flip-flops which operate in synchronism with external clock pulses are known as

Select one:

- ☐ a. None of these
- ☐ b. Either of the above
- ☒ c. Synchronous flip-flop ✓
- ☐ d. Asynchronous flip-flop

The correct answer is: Synchronous flip-flop

Question 19

Correct

Mark 1.50 out of 1.50

Flag question

How many flip-flop are needed to divide the input frequency by 64 ?

Select one:

- ☐ a. 8
- ☒ b. 6 ✓
- ☐ c. 5
- ☐ d. 2

The correct answer is: 6

Question 20

Correct

Mark 1.50 out of
1.50

Flag question

How many lines the truth table for a four-input NOR gate would contain to cover all possible input combinations ?

Select one:

- ☐ a. 12
- ☒ b. 16 ✓
- ☐ c. 8
- ☐ d. 4

The correct answer is: 16

Question 21

Incorrect

Mark 0.00 out of
1.50

Flag question

The master slave JK lip-flop is effectively a combination of

Select one:

- ☒ a. Two D flip-flops ✗
- ☐ b. A T flip-flop and a D flip-flop
- ☐ c. A SR flip-flop and a T flip-flop
- ☐ d. An SR flip-flop and a D flip-flop

The correct answer is: A SR flip-flop and a T flip-flop

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Question 22

Incorrect

Mark 0.00 out of
1.50

Flag question

The number of full and half-adders required to add 16-bit numbers is

Select one:

- ☒ a. 4 half-adders, 12 full-adders ✖
- ☐ b. 1 half-adder, 15 full-adders
- ☐ c. half-adders, 8 full-adders
- ☐ d. 16 half-adders, 0 full-adders

The correct answer is: 1 half-adder, 15 full-adders

Question 23

Correct

Mark 1.50 out of
1.50

Flag question

What logic function is produced by adding an inverter to the output of an AND gate ?

Select one:

- ☐ a. NOR
- ☐ b. XOR
- ☐ c. OR
- ☒ d. NAND ✔

The correct answer is: NAND

Question 24

Correct

Mark 1.50 out of
1.50

Flag question

Given $F1(A,B,C,D) = \text{SOP}(0, 2, 4, 8)$, $F2(A,B,C,D) = \text{SOP}(1, 2, 3, 4, 15)$ and $F3 = F1 + F2$ then:

Select one:

- ☐ a. $F3(A,B,C,D) = \text{SOP}(0, 1, 2, 3, 4)$
- ☒ b. $F3(A,B,C,D) = \text{SOP}(0, 1, 2, 3, 4, 8, 15)$ ✓
- ☐ c. $F3(A,B,C,D) = \text{SOP}(0, 1, 2, 3, 4, 8)$
- ☐ d. $F3(A,B,C,D) = \text{SOP}(2, 4)$

The correct answer is: $F3(A,B,C,D) = \text{SOP}(0, 1, 2, 3, 4, 8, 15)$

Question 25

Incorrect

Mark 0.00 out of
1.50

Flag question

When an inverter is placed between both inputs of an SR flip-flop, then resulting flip-flop is

Select one:

- ☐ a. JK flip-flop
- ☒ b. Master slave JK flip-flop ✗
- ☐ c. SR flip-flop
- ☐ d. D flip-flop

The correct answer is: D flip-flop

Question 26

Incorrect

Mark 0.00 out of
1.50

Flag question

In a positive edge triggered JK flip-flop, a low J and low K produces

Select one:

- ☐ a. None of these
- ☐ b. No change
- ☐ c. Low state
- ☒ d. High state ❌

The correct answer is: No change

Question 27

Incorrect

Mark 0.00 out of
1.50

Flag question

A stable multivibrator is used as

Select one:

- ☐ a. Demultiplexer
- ☐ b. Voltage to frequency converter
- ☒ c. Frequency to voltage converter ❌
- ☐ d. Comparator circuit

The correct answer is: Comparator circuit

Question 28

Correct

Mark 1.50 out of 1.50

Flag question

The output of NOR gate is

Select one:

- ☐ a. High if only of its inputs is low
- ☐ b. High if all of its inputs are high
- ☐ c. Low if all of its inputs are low
- ☒ d. High if all of its inputs are low ✓

The correct answer is: High if all of its inputs are low

Question 29

Correct

Mark 1.50 out of 1.50

Flag question

How many minimal numbers of basic logic gates (NOT, AND, OR) need to represent the Boolean function ?

Select one:

- ☒ a. One NOT, one OR. ✓
- ☐ b. One NOT, one AND.
- ☐ c. Two NOTs, one OR, one AND.
- ☐ d. Two NOT, one OR.

The correct answer is: One NOT, one OR.

Question 30

Incorrect

Mark 0.00 out of
1.50

Flag question

Adders:

Select one:

- ☒ a. is called so because a full adder involves two half-adders 
- ☐ b. needs two input and generates two output
- ☐ c. All of these
- ☐ d. adds 2 bits

The correct answer is: All of these



QUIZ NAVIGATION

1	2	3	4	5	6	7	8	9
✓		✓	✓					
10	11	12	13	14	15	16	17	18
✓	✓	✓	✓	✓				✓
19	20	21	22	23	24	25	26	27
✓	✓			✓	✓	✓	✓	✓
28	29	30						
		✓						

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Started on Tuesday, 19 May 2020, 1:00 PM

State Finished

Completed on Tuesday, 19 May 2020, 1:40 PM

Time taken 40 mins

Grade 29.50 out of 50.00 (59%)

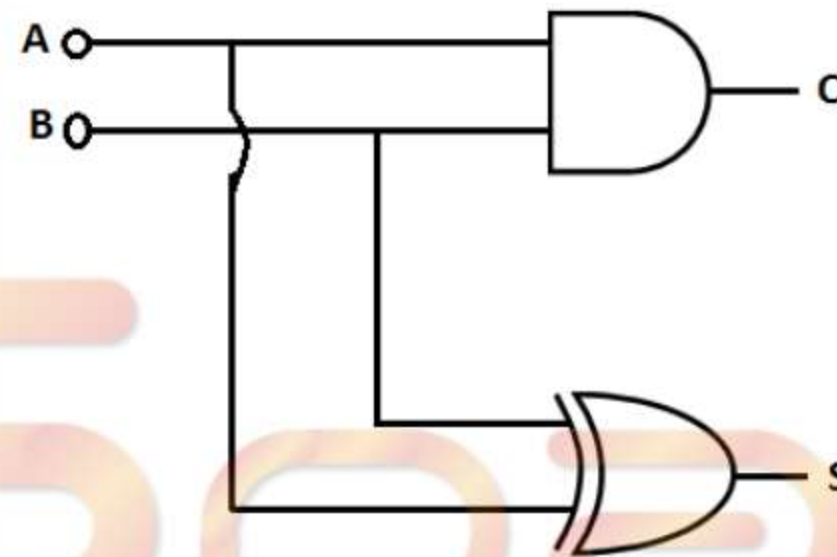
Question 1

Correct

Mark 2.50 out of 2.50

Flag question

For the circuit shown for $AB = 00$, $AB = 01$, C , S values respectively are



Select one:

- ☐ a. 0, 1 and 0, 0
- ☐ b. 0, 0 and 1, 0
- ☒ c. 0, 0 and 0, 1 ✓
- ☐ d. 1, 0 and 0, 0

Your answer is correct.

The correct answer is: 0, 0 and 0, 1

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Question 2

Incorrect

Mark 0.00 out of
1.50

Flag question

For which of the following flip-flops, the output is clearly defined for all combinations of two inputs ?

Select one:

- ☒ a. D flip-flop ❌
- ☐ b. J-K flip-flop
- ☐ c. Q type flip-flop
- ☐ d. R-S flip-flop

The correct answer is: J-K flip-flop

Question 3

Correct

Mark 1.50 out of
1.50

Flag question

The output of NOR gate is

Select one:

- ☐ a. High if all of its inputs are high
- ☐ b. Low if all of its inputs are low
- ☒ c. High if all of its inputs are low ✔️
- ☐ d. High if only of its inputs is low

The correct answer is: High if all of its inputs are low

Question 4

Correct

Mark 1.50 out of
1.50

Flag question

The digital multiplexer is basically a combination logic circuit to perform the operation

Select one:

- ☒ a. AND-OR ✔️
- ☐ b. AND-AND
- ☐ c. OR-AND
- ☐ d. OR-OR

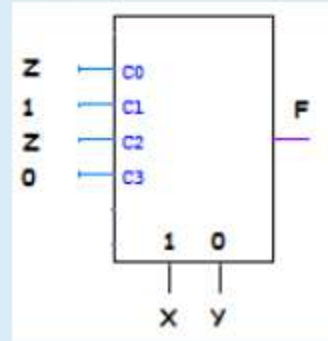
Question 5

Incorrect

Mark 0.00 out of
1.50

Flag question

The function that can be implemented from the following device is:

 $F(X,Y,Z) = \dots\dots\dots$ 

Select one:

- ☐ a. $F = X'Y + Y'Z$
- ☐ b. $F = 0$
- ☐ c. $F = X'Y'Z + XY'Z$
- ☒ d. $F = 1$ ✖

The correct answer is: $F = X'Y + Y'Z$ **Question 6**

Incorrect

Mark 0.00 out of
1.50

Flag question

The EXCLUSIVE NOR gate is equivalent to which gate followed by an inverter ?

Select one:

- ☐ a. AND
- ☐ b. NAND
- ☒ c. OR ✖
- ☐ d. XOR

The correct answer is: XOR

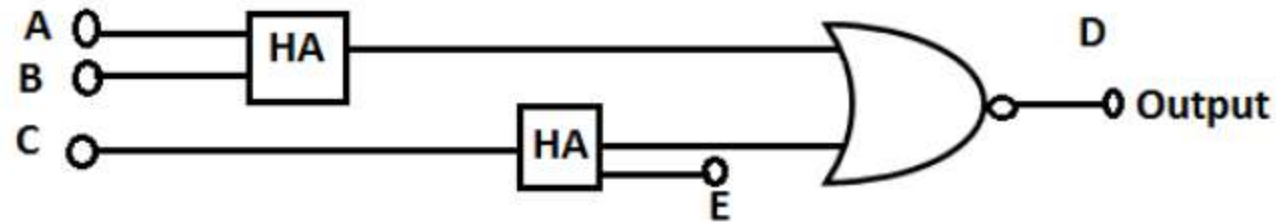
Question 7

Incorrect

Mark 0.00 out of
2.50

Flag question

The circuit shown below is the



Select one:

- ☐ a. Full subtractor
- ☐ b. Full adder
- ☐ c. None of these
- ☒ d. Parity checker ✖

Your answer is incorrect.

The correct answer is: Full adder

Question 8

Incorrect

Mark 0.00 out of
1.50

Flag question

A combinational logic circuit which generates a particular binary word or number is

Select one:

- ☒ a. Encoder ✖
- ☐ b. Demultiplexer
- ☐ c. Multiplexer
- ☐ d. Decoder

The correct answer is: Decoder

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Question 9

Incorrect

Mark 0.00 out of 1.50

Flag question

Which of the following **adders** can add two binary numbers at a time ?

Select one:

- ☐ a. Parallel adder
- ☐ b. Carry-look-ahead adder
- ☐ c. Carry-save-adder
- ☒ d. Full adder **✗**

The correct answer is: Carry-save-adder

Question 10

Correct

Mark 2.50 out of 2.50

Flag question

The combinational circuit given below is implemented with two NAND gates. To which of the following individual gates is its equivalent?



Select one:

- ☐ a. XOR
- ☐ b. OR
- ☐ c. NOT
- ☒ d. AND **✓**

Your answer is correct.

The correct answer is: AND

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Question 11

Correct

Mark 1.50 out of 1.50

Flag question

Which of the following expressions is not equivalent to X' ?

Select one:

- ☐ a. $X \text{ NAND } 1$
- ☐ b. $X \text{ NOR } X$
- ☒ c. $X \text{ NOR } 1$ ✓
- ☐ d. $X \text{ NAND } X$

The correct answer is: $X \text{ NOR } 1$

Question 12

Correct

Mark 1.50 out of 1.50

Flag question

In a positive edge triggered JK flip-flop, a low J and low K produces

Select one:

- ☒ a. No change ✓
- ☐ b. High state
- ☐ c. Low state
- ☐ d. None of these

The correct answer is: No change

Question 13

Correct

Mark 1.50 out of 1.50

Flag question

How many lines the truth table for a four-input NOR gate would contain to cover all possible input combinations ?

Select one:

- ☐ a. 8
- ☐ b. 4
- ☒ c. 16 ✓
- ☐ d. 12

Question 14

Correct

Mark 1.50 out of 1.50

Flag question

What logic function is produced by adding an inverter to the output of an AND gate ?

Select one:

- ☐ a. XOR
- ☒ b. NAND ✓
- ☐ c. NOR
- ☐ d. OR

The correct answer is: NAND

Question 15

Incorrect

Mark 0.00 out of 1.50

Flag question

Given $F1(A,B,C,D) = \text{SOP}(0, 2, 4, 8)$, $F2(A,B,C,D) = \text{SOP}(1, 2, 3, 4, 15)$ and $F3 = F1 + F2$ then:

Select one:

- ☐ a. $F3(A,B,C,D) = \text{S}(0, 1, 2, 3, 4, 8, 15)$
- ☐ b. $F3(A,B,C,D) = \text{S}(0, 1, 2, 3, 4)$
- ☒ c. $F3(A,B,C,D) = \text{S}(2, 4)$ ✗
- ☐ d. $F3(A,B,C,D) = \text{S}(0, 1, 2, 3, 4, 8)$

The correct answer is: $F3(A,B,C,D) = \text{S}(0, 1, 2, 3, 4, 8, 15)$

Question 16

Incorrect

Mark 0.00 out of 1.50

Flag question

When an inverter is placed between both inputs of an SR flip-flop, then resulting flip-flop is

Select one:

- ☐ a. SR flip-flop
- ☐ b. D flip-flop
- ☒ c. Master slave JK flip-flop ✗
- ☐ d. JK flip-flop

Question 17

Incorrect

Mark 0.00 out of 1.50

Flag question

To implement the Boolean function $F = A \oplus B \oplus C$ we can use a 3-to-8 decoder and connect:

Select one:

- ☒ a. 3 different lines from its output to an OR gate ✗
- ☐ b. 5 different lines from its output to an OR gate
- ☐ c. 6 different lines from its output to an OR gate
- ☐ d. 4 different lines from its output to an OR gate

The correct answer is: 4 different lines from its output to an OR gate

Question 18

Correct

Mark 1.50 out of 1.50

Flag question

If the input J is connected through K input of J-K, then flip-flop will behave as a

Select one:

- ☒ a. T type flip-flop ✓
- ☐ b. Master slave JK flip-flop
- ☐ c. S-R flip-flop
- ☐ d. D type flip-flop

The correct answer is: T type flip-flop

Question 19

Correct

Mark 2.50 out of 2.50

Flag question

Which of the following gates are added to the inputs of the OR gate to convert it to the NAND gate ?

Select one:

- ☐ a. AND
- ☒ b. NOT ✓
- ☐ c. XOR
- ☐ d. OR

Question 20

Correct

Mark 1.50 out of 1.50

Flag question

How many flip-flop are needed to divide the input frequency by 64 ?

Select one:

- ☒ a. 6 ✓
- ☐ b. 5
- ☐ c. 2
- ☐ d. 8

The correct answer is: 6

Question 21

Incorrect

Mark 0.00 out of 1.50

Flag question

What is the minimum number of 2 input NAND gates required to implement the function $(x'+y')(z+w)$:

Select one:

- ☐ a. 3
- ☐ b. 4
- ☒ c. 5 ✗
- ☐ d. 6

The correct answer is: 4

Question 22

Incorrect

Mark 0.00 out of 1.50

Flag question

A demultiplexer is used to

Select one:

- ☐ a. Route the data from single input to one of many outputs
- ☐ b. Both route the data and perform serial to parallel
- ☒ c. Perform serial to parallel conversion ✗
- ☐ d. Select data from several inputs and route it to single output

Question 23

Correct

Mark 1.50 out of
1.50

Flag question

The **flip-flops** which operate in synchronism with external clock pulses are known as

Select one:

- ☐ a. None of these
- ☐ b. Either of the above
- ☒ c. Synchronous flip-flop ✓
- ☐ d. Asynchronous flip-flop

The correct answer is: Synchronous flip-flop

Question 24

Correct

Mark 1.50 out of
1.50

Flag question

Adders:

Select one:

- ☒ a. All of these ✓
- ☐ b. needs two input and generates two output
- ☐ c. is called so because a full adder involves two half-adders
- ☐ d. adds 2 bits

The correct answer is: All of these

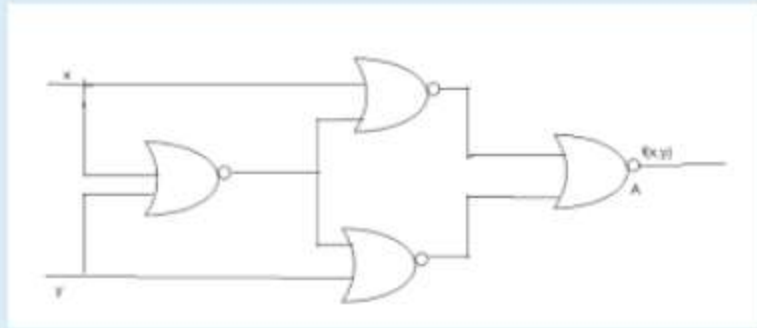
Question 25

Correct

Mark 2.50 out of 2.50

Flag question

Identify the logic function performed by the circuit shown in the given figure



Select one:

- ☐ a. NOR
- ☒ b. Exclusive NOR ✓
- ☐ c. NAND
- ☐ d. Exclusive OR

Your answer is correct.

The correct answer is: Exclusive NOR

Question 26

Correct

Mark 1.50 out of 1.50

Flag question

The function of a multiplexer is

Select one:

- ☐ a. to decode information
- ☐ b. to transit data on N lines
- ☒ c. to select 1 out of N input data sources and to transmit it to single channel ✓
- ☐ d. to perform serial to parallel conversion

The correct answer is: to select 1 out of N input data sources and to transmit it to single channel

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Question 27

Correct

Mark 1.50 out of 1.50

Flag question

What is the largest number of data inputs which a data selector with two control inputs can have ?

Select one:

- ☐ a. 2
- ☐ b. 8
- ☐ c. 16
- ☒ d. 4 ✓

The correct answer is: 4

Question 28

Incorrect

Mark 0.00 out of 1.50

Flag question

How many minimal numbers of basic logic gates (NOT, AND, OR) need to represent the Boolean function ?

Select one:

- ☐ a. Two NOTs, one OR, one AND.
- ☒ b. One NOT, one AND. ✗
- ☐ c. One NOT, one OR.
- ☐ d. Two NOT, one OR.

The correct answer is: One NOT, one OR.

Question 29

Incorrect

Mark 0.00 out of
1.50

Flag question

The number of full and half-adders required to add 16-bit numbers is

Select one:

- ☒ a. 16 half-adders, 0 full-adders ❌
- ☐ b. 4 half-adders, 12 full-adders
- ☐ c. half-adders, 8 full-adders
- ☐ d. 1 half-adder, 15 full-adders

The correct answer is: 1 half-adder, 15 full-adders

Question 30

Correct

Mark 1.50 out of
1.50

Flag question

An OR gate can be imagined as

Select one:

- ☒ a. Switches connected in parallel ✔️
- ☐ b. None of these
- ☐ c. Switches connected in series
- ☐ d. MOS transistors connected in series

The correct answer is: Switches connected in parallel